

Daniel Robles, Ph.D.

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EDUCATION

University of Alberta, Edmonton, AB, Canada

Ph.D., Psychology (2023)

Concentration: Behavioral, Systems, and Cognitive Neuroscience

The City College of New York, New York, NY

M.A., Psychology (2017)

The City College of New York, New York, NY

B.A., Psychology (2011)

RESEARCH EXPERIENCE

Baker Laboratory for Cognitive Neuroimaging and Stimulation, Center for Molecular and Behavioral Neuroscience, Rutgers University

Postdoctoral Associate (July 2023–present)

Principal Investigator: Travis Baker

AppLab, Department of Psychology, University of Alberta

Graduate Research Assistant (September 2017–February 2023)

Principal Investigator: Kyle Mathewson

Child Development Study, Center for Child Well-Being & Development, University of Zurich

Research Member (September 2019–present)

Principal Investigator: Guilherme Lichand

Sophie Davis School of Biomedical Education, CUNY

Research Assistant (January 2017–August 2017)

Principal Investigator: Philip Smith

Melara Lab, Department of Psychology, The City College of New York

Graduate Research Assistant/Coordinator (2014–2017)

Primary Investigator: Robert Melara

Translational Research Training in Addictions for Racial/Ethnic Minority Students at City College of New York and Columbia University Medical Center.

Research Fellow (2014–2015)

Primary Investigator: Denise Hien

Department of Psychology, The City College of New York

Research Assistant (2013–2014)
Primary Investigator: Lesia Ruglass

HONORS AND AWARDS

- Canadian Psychological Association. Certificate of Academic Excellence 2023
- University of Alberta Graduate Student Engagement Scholarship (2022)
- Mary Louise Imrie Graduate Student Award (2021)
- TRACC Alumni Travel Award (2021)
- University of Alberta Graduate Students Teaching Award (2020)
- Graduate Students' Association Academic Travel Grant (2018)
- University of Alberta Doctoral Recruitment Scholarship (2017)
- Psi Chi, The International Honor Society in Psychology (2011)

TEACHING EXPERIENCE

Lecturer, Rutgers University (2026–present)

- Foundations III: Introduction to Cognitive Neuroscience (Spring 2025, Fall 2026)
- Trends in Neuroscience (Spring 2026)

Adjunct Lecturer, The City College of New York (2025–present)

- Psychology 25400: Mind, Brain, and Experience (Fall 2026, Spring 2026, Fall 2025)

Guest Lecturer, Rutgers University (2023–2025)

- Introduction to Neuroscience (Fall 2025)
- Foundations of Neuroscience and Behavior (Fall 2023, Fall 2024)

Assistant Lecturer, University of Alberta (2022)

- Psych 275: Brain and Behavior (Fall 2022)
- Psych 275: Brain and Behavior (Spring 2022)
- Psych 105: Basic Psychological Processes (Winter 2022)

Teaching Assistant, University of Alberta (September 2017–2021)

Adjunct Lecturer, The City College of New York (2015–2017)

- Psychology 35600: Adolescence & Youth (Spring 2015, Spring 2016, Summer 2016, Spring 2017, Summer 2017)
- Psychology 25300: Thinking, Knowing, and Remembering (Spring 2016, Fall 2016, Spring 2017)
- Psychology 37300: Neuropsychology (Spring 2017)
- Psychology 10200: Psychology in the Modern World (Fall 2015)
- Psychology 35000: Treatment of Substance Abuse I (Summer 2015)

Teaching Assistant, The City College of New York (2013–2015)

PEER-REVIEWED PUBLICATIONS

Shevorykin, A., Hyland, B. M., **Robles, D.**, Ji, M., Vantucci, D., Bensch, L., ... Sheffer, C. E. (2024). Tobacco use, trauma exposure and PTSD: a systematic review. *Health Psychology Review*, 18(4), 649–680.
<https://doi.org/10.1080/17437199.2024.2330896>

Mathewson KE, Kuziek JP, Scanlon JEM, **Robles D.** The moving wave: Applications of the mobile EEG approach to study human attention. *Psychophysiology*, 61(9), e14603. <https://doi.org/10.1111/psyp.14603>

Baker, T. E., & **Robles, D.** (2023). Theta burst stimulation of the hyperdirect pathway boosts inhibitory control and reduces craving and smoking in nicotine-dependent adults. *Biological Psychiatry: Cognitive Neuroscience and Neuroimaging*, 8(11), 1072-1074. <https://doi.org/10.1016/j.bpsc.2023.09.007>

Robles, D., Kuziek, J. W. P., Wlasitz, N. A., Bartlett, N. T., Hurd, P. L., & Mathewson, K. E. (2021). EEG in motion: Using an oddball task to explore motor interference in active skateboarding. *European Journal of Neuroscience*, n/a(n/a).
<https://doi.org/https://doi.org/10.1111/ejn.15163>

Leal Neto, O., Haenni, S., Phuka, J., Ozella, L., Paolotti, D., Cattuto, C., **Robles, D.**, & Lichand, G. (2021). Combining wearable devices and mobile surveys to study child and youth development in Malawi: Implementation study of a multimodal approach. *JMIR Public Health and Surveillance*, 7(3), e23154.
<https://doi.org/10.2196/23154>

PREPRINTS

Robles, D., Kuziek, J. W., Lai, J., Mazumder, R., Scanlon, J. E., & Mathewson, K. E. (2022). Surrounding traffic matters: Increases in traffic volume are related to changes in EEG rhythms in urban cyclists. *bioRxiv*.
<https://doi.org/10.1101/2022.05.27.493782>

Lichand, G., Leal Neto, O., Phuka, J., Chipojola, R., Laher, B., Enlow, M. B., **Robles, D.**, ... & Carneiro, P. M. (2022). The early childhood development replication crisis, and how wearable technologies could help overcome it. *SSRN*.
<https://dx.doi.org/10.2139/ssrn.4162049>

MANUSCRIPTS IN PREPARATION

Robles, D., Zhang, E., Bharti, B.H., and Baker, T.E. (in preparation). Dissociable TMS Protocols for Bidirectional Control of Reward Processing in Smokers.

INVITED TALKS

Robles, D. (2025, January). Comparing the effects of 3600 pulses of 10-Hz TMS and cTBS on reward functioning in smokers. Invited oral presentation at the NIH T-34 Winter Training Institute (WTI), New York, NY.

Robles, D. (2021, April). Mobile EEG: Contributions and challenges to the study of human attention. Invited oral presentation at the Department of Psychology Weekly Cognitive Seminar, University of Alberta, Edmonton, AB, Canada.

Robles, D. (2021, March). Portable EEG and the solution for portable neuroscience. Invited oral presentation at the Brown Bag series, Center for Child Well-Being and Development, University of Zurich, Switzerland.

Robles, D. (2018, November). Getting on TRACC with a research career. Invited panel presentation for the Translational Research Training in Addictions for Racial/Ethnic Minorities (TRACC), The City College of New York and Columbia University Medical Center, New York, NY.

SELECTED PRESENTATIONS

Robles, D., Baker, T. (2026, May). Exploring the effects of bidirectional TMS stimulation in cognitive control during reinforcement learning. Poster presentation at the Rutgers Symposium on the Intersection of Cognitive, Sensory, and Affective Neuroscience, Union County, NJ.

Robles, D. (2025, November). Differential effects of transcranial magnetic stimulation protocols on reward processing in abstinent smokers: An EEG study of dlPFC stimulation. Oral presentation at the 33rd Annual CMBN Mini-Symposium, Newark, NJ.

Gao, Y., Güth, M., Headley, D., **Robles, D.**, Zhang, E., Baker, T. (2025, April). Real-time modulation of reinforcement learning using closed-loop TMS-EEG. Poster presented at the Cognitive Neuroscience Annual Meeting, Boston, MA.

Gao, Y., **Robles, D.**, Zhang, E., Güth, M., Headley, D., Baker, T. (2024, August). Closed-loop TMS-EEG modulation of reward-related ERPs. Poster presented at the NYC Neuromodulation Conference, New York, NY.

Baker, T. E., Reid, A., Cole, M. W., Zhang, E., **Robles, D.**, Mill, R. D., Lalta, N. D., Güth, M. (2024). Quantifying and modulating reward circuit dysfunction in substance use disorders using a combination of human electrophysiology, neuroimaging, and robot-assisted TMS. Presented at the 6th International Brain Stimulation Conference, Kobe, Japan.

Robles, D., Zhang, E., Lalta, N., Baker, T. (2024). Comparing the effects of 3600 pulses of 10-Hz TMS and cTBS on reward functioning in smokers. Poster presented at

the NYC Neuromodulation Conference, New York, NY.

Lalta, N. D., Zhang, E., **Robles, D.**, Güth, M., Mill, R. D., Reid, A., Cole, M. W., Baker, T. E. (2024). Enhancing cognitive control functioning in nicotine users using individualized resting-state functional connectivity TMS targets. Poster presented at the NYC Neuromodulation Conference, New York, NY.

Zhang, E., Lalta, N. D., **Robles, D.**, Güth, M. R., Reid, A., Mill, R. D., Cole, M. W., Baker, T. E. (2024). Determining the optimal TMS image-based target to modulate reward and control functioning in nicotine users. Poster presented at the NYC Neuromodulation Conference, New York, NY.

Robles, D., Mathewson, K. E. (2022, September). Increases in traffic volume are associated with measurable changes in EEG in urban cycling lanes. Poster presented at the 62nd Annual Meeting of the Society for Psychophysiological Research, Vancouver, BC, Canada.

Sheldon, S., **Robles, D.**, Mathewson, K. E. (2021, October). Decoding covert attention on an orientation perception task from EEG alpha activity. Poster presented at the 61st Annual Meeting of the Society for Psychophysiological Research.

Robles, D. (2021, August). EEG on the go: Using novel approaches to study human attention in real-world environments. Oral presentation at the 2021 Sapiens Labs Virtual Symposium on EEG: Analytical Approaches and Applications.

Robles, D. (2021, April). EEG in motion: Using an oddball task to explore motor interference in active skateboarding. Oral presentation at the 2021 Royce-Harder Research Conference, Edmonton, AB, Canada.

Robles, D., Kuziek, J., Mazumder, R., Lai, J., Mathewson, K. E. (2020, October). Differences in traffic conditions are related to N1 amplitude changes during cycling. Poster presented at the 60th Annual Meeting of the Society for Psychophysiological Research.

Robles, D., Wlasitz, N., Bartlett, N., Hurd, P., Mathewson, K. E. (2019, May). Brainwaves on a skateboard: Exploring laterality in skateboarders using electrophysiological and behavioral measures. Poster presented at the 37th Banff Annual Seminar in Cognitive Science, Banff, AB, Canada.

Robles, D., Sheldon, S., Mathewson, K. E. (2018, October). Feedback error-related negativity as a control signal for the attention system. Poster presented at the 58th Annual Meeting of the Society for Psychophysiological Research, Quebec City, Quebec, Canada.

Robles, D., Mathewson, K. E. (2018, April). Feedback error-related negativity as a control signal for the attention system. Oral presentation at the 2018

Royce-Harder Research Conference, Edmonton, AB, Canada.

Robles, D., Melara, R. D. (2018, February). Differences in alpha-related inhibitory control between smokers and ex-smokers. Poster presented at the 13th Covenant Health Research Day, Edmonton, AB, Canada.

Robles, D., Shevorykin, A., Melara, R. D. (2017, March). Attentional differences in the presence of cigarette cues among smokers, ex-smokers, and non-smokers. Poster presented at the 2017 Joseph R. Royce Conference, Edmonton, AB, Canada.

Shevorykin, A., **Robles, D.,** Baig, A., Dambreville, N., Ruglass, L. M., Melara, R. D. (2017, March). *The Effect of Childhood Abuse in Cigarette Smoking Behavior*. Poster to be presented at the 23rd Annual Meeting of the Society for Research on Nicotine and Tobacco, Florence, Italy.

Shevorykin, A., Baig, A., **Robles, D.,** Ruglass, L. M., Melara, R. D. (2016, April). Attentional bias in individuals with comorbid cannabis use disorders and tobacco use. Poster presented at the 23rd Annual Meeting of the Cognitive Neuroscience Society, New York, NY.

Robles, D., Shevorykin, A., Bhatti, M., Ovalles, M., Cabrera, J., Melara, R. D. (2016, March). Attentional differences in the presence of cigarette cues among smokers, ex-smokers, and non-smokers. Poster presented at the 22nd Annual Meeting of the Society for Research on Nicotine and Tobacco, Chicago, Illinois.

Shevorykin, A., Ruglass, L. M., Tsoi, F., Baig, A., Bibi, R., John, G., **Robles, D.,** Melara, R. D. (2015, May). Psychological factors associated with severity of cannabis use disorder. Poster presented at the 27th Annual Convention of the Association for Psychological Science, New York, NY.

Robles, D., Shevorykin, A., Melara, R. D. (2015, April). ADHD symptoms and nicotine dependence in urban populations. Poster presented at the 4th International Congress of Dual Disorders, Barcelona, Spain.

Shevorykin, A., Ruglass, L. M., **Robles, D.,** Tsoi, F., Melara, R. D. (2015, April). The associations between mood, craving level, and symptom severity in people with marijuana use disorder. Poster presented at the 4th International Congress of Dual Disorders, Barcelona, Spain.

Shevorykin, A., **Robles, D.,** Moses, J., Melara, R. D. (2014, November). Cognitive difficulties in an urban population of cigarette smokers. Poster presented at the 44th Annual Meeting of the Society for Neuroscience, Washington, DC.

VOLUNTEER SERVICES

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- Poster Judge, CMBN Mini-Symposium, Rutgers University, Newark, NJ (2023)
 - Administrative committee, Graduate Psychology Association, Edmonton, AB (2018–2022)
 - Royce-Harder Conference Organizing Committee, Edmonton, AB (2022)
 - Co-mentor and Discussion Panel Leader, WISEST SET Conference (2019)

PROFESSIONAL AFFILIATIONS

- Society for Psychophysiological Research (SPR)
- Cognitive Neuroscience Society (CNS)
- Society for Research on Nicotine and Tobacco (SRNT)

PEER REVIEW ACTIVITY

- Journal of Cognitive Neuroscience
- European Journal of Neuroscience
- Developmental Cognitive Neuroscience
- Current Protocols

OPEN SCIENCE

GitHub repositories with open-source analysis code and data pipelines:

- mobile_ersp: Mobile EEG time-frequency analysis pipeline (https://github.com/d-robles/mobile_ersp)
- skate_backup: Skateboarding EEG experiment scripts and analysis (https://github.com/d-robles/skate_backup)

MEDIA & PUBLIC ENGAGEMENT

- Malawi EEG work featured on the University of Alberta's website. [University of Alberta \(2020\)](#)

TECHNICAL SKILLS

Neuroimaging & Neuromodulation: EEG acquisition and analysis, event-related potentials (ERPs), TMS (10 Hz, cTBS, iTBS protocols), TMS-EEG, robot-assisted TMS, individualized TMS targeting (coil angle/distance optimization), SimNIBS, MRI processing (cortical thickness, diffusion imaging), mobile/ambulatory EEG

Programming & Data Analysis: MATLAB, Python, R, STATA, SPSS, JASP, computational modeling (Q-learning)

Software & Platforms: Microsoft Office Suite, HTML, web design, Windows, macOS,

Linux

LANGUAGE SKILLS

- English — Full professional proficiency (read, write, speak, understand)
- Spanish — Full professional proficiency (read, write, speak, understand)
- Portuguese — Professional working proficiency (read, speak, understand; limited writing)